

A.O. SMITH

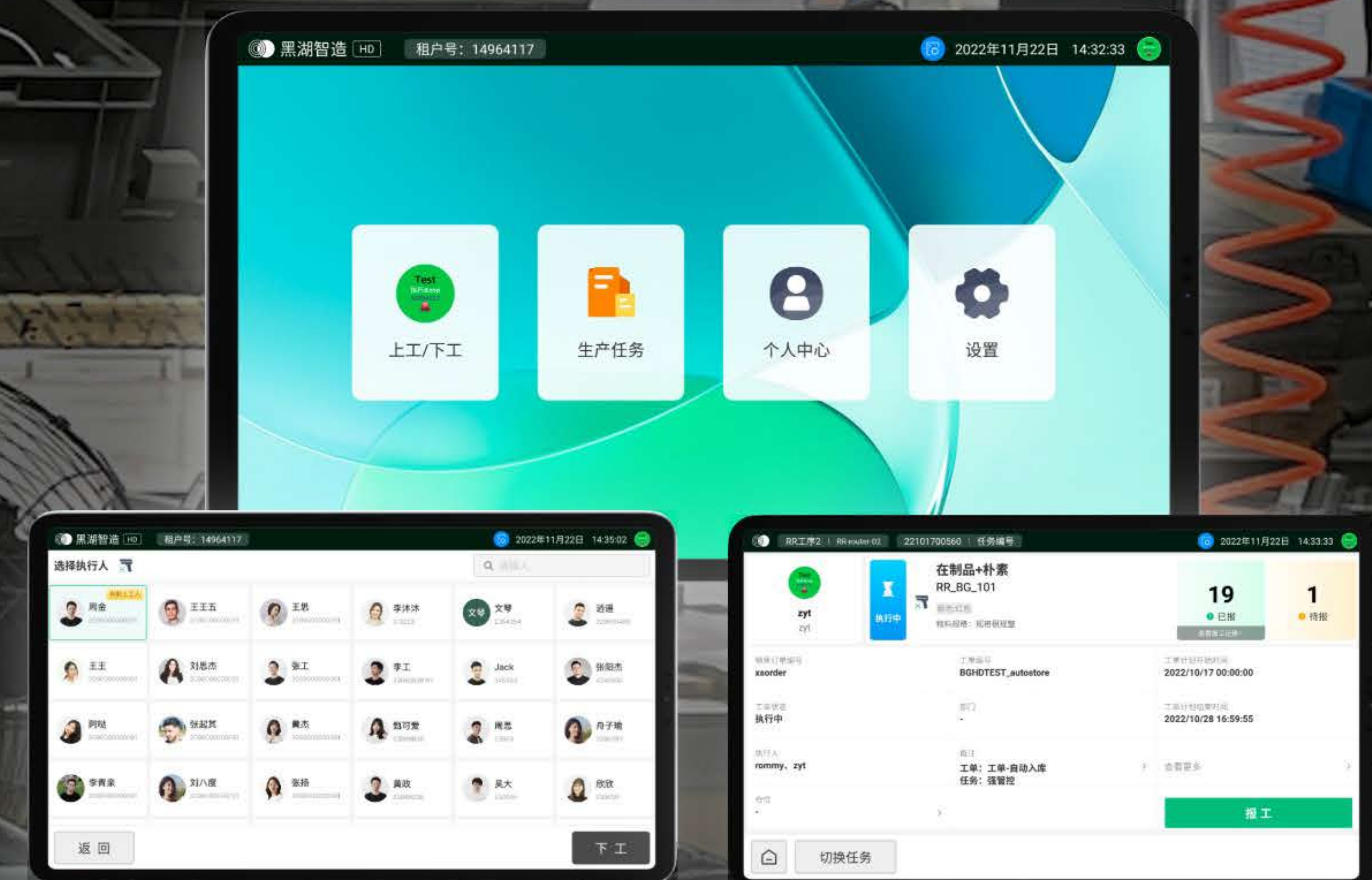
~~BLIM~~  BLIM-HD

Based on the horizontal screen hardware UX design of the Manufacturing Execution System for workers in a fixed workstation scenario

Co-Design / UX Design

 ShangHai · 2022

 Black Lake Technologies × 



Background



Blacklake Intelligent Manufacturing (BLIM) (worker side)

is an App that records worker material input & worker working hours, material outbound, inbound, and transfer.

- 2016 ● BLIM has used mobile phones as the hardware terminal
- 2021 ● A.O. Smith subscribed to BLIM service
- 2022 ● BLIM was complained from A.O. Smith for the reason of **Overload of information and complex operations.**



Min (GM of Intelligent Manufacturing (IM) Business Line) wanted to optimize the interaction design of **BLIM**, **but lacked competing products** on the market, and no simplified work hours recording system to refer to.

This opportunity led me to join BL and be assigned to this project in 2022.

MingLei

TeamLeader of front-end Engineer

Rong

Customer Service for A.O

Eason

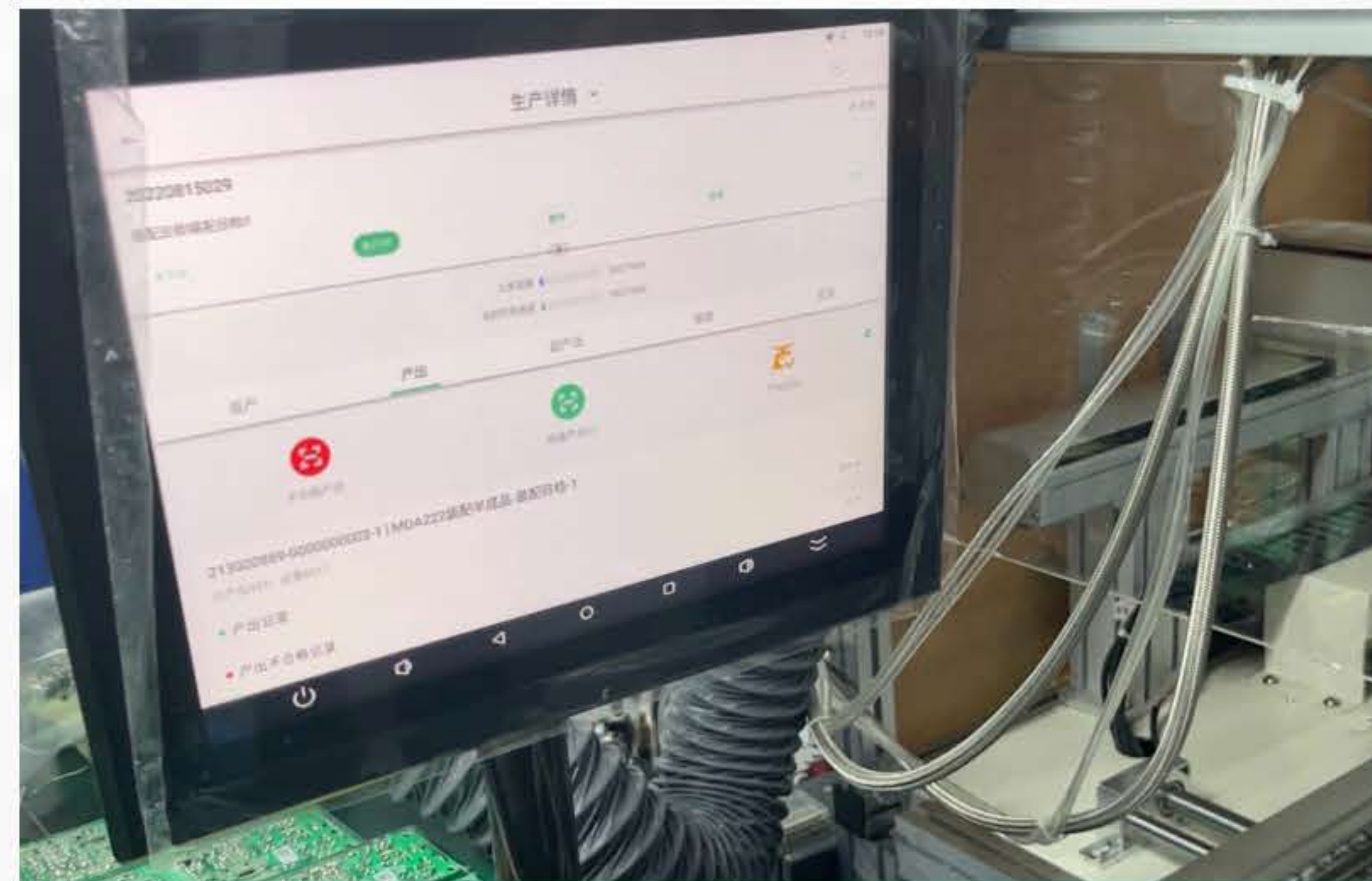
Product Manager

Me: Wenzhe Xu
UX designer

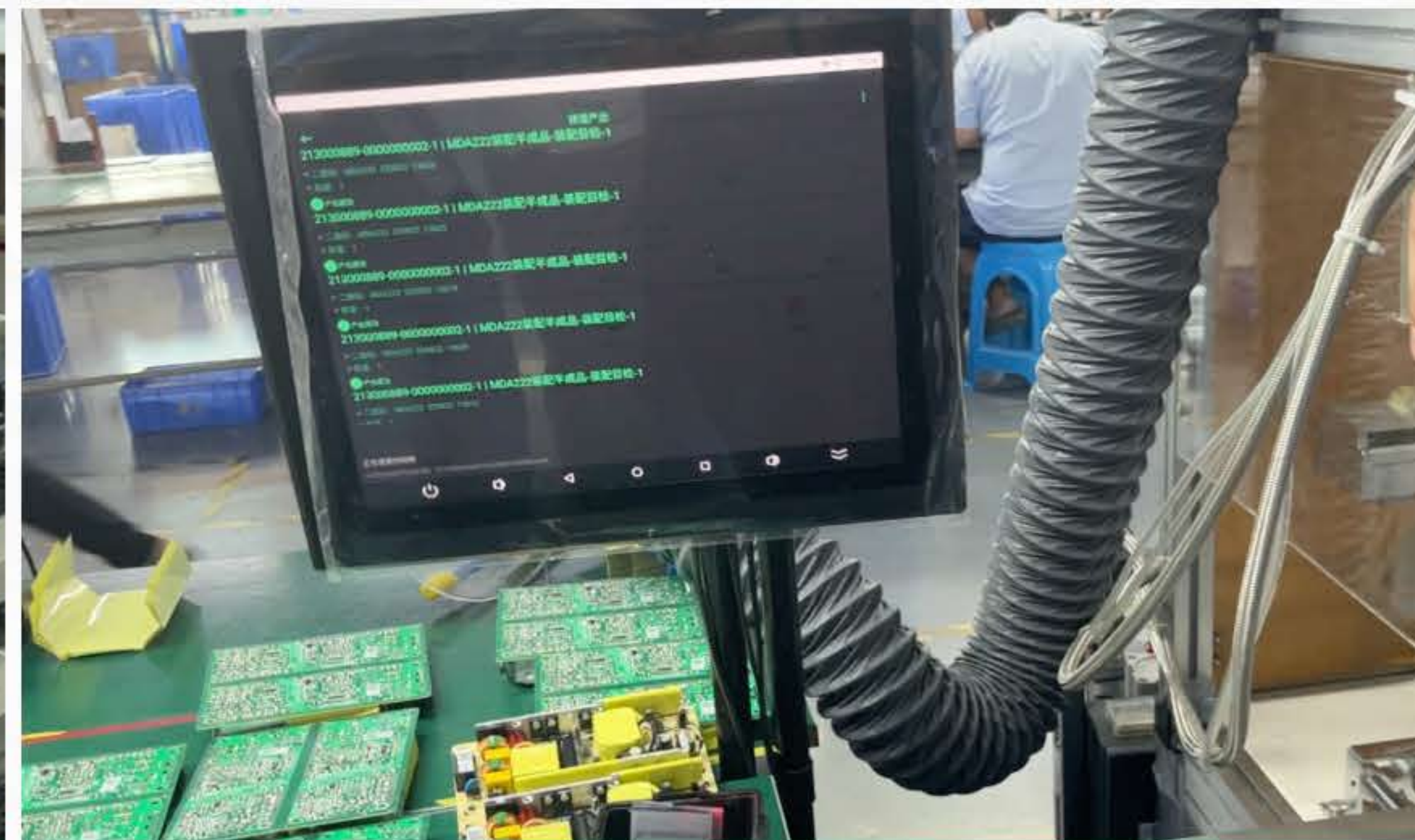
Why A.O smiths' worker complain?

1. Field research & Structured Observation

Replace the mobile phone with a pad



Replace the mobile phone with a pad



Replace the mobile phone with a pad

Fixed workstations



fixed workstations



fixed workstations

Finding 1

The factory voluntarily moved the BLIM, which should have been operated via mobile phone, to a PAD for operation;

Finding 2

Workers were assigned to fixed processes and assembled at fixed workstations, without moving around while performing their tasks.

*BL3.0的初衷是基于手机的移动性和便利性做的系统设计。
In scenarios where workers are mobile, mobile phones are convenient to carry, but the information that a screen can hold is limited, so readability is not an advantage.*

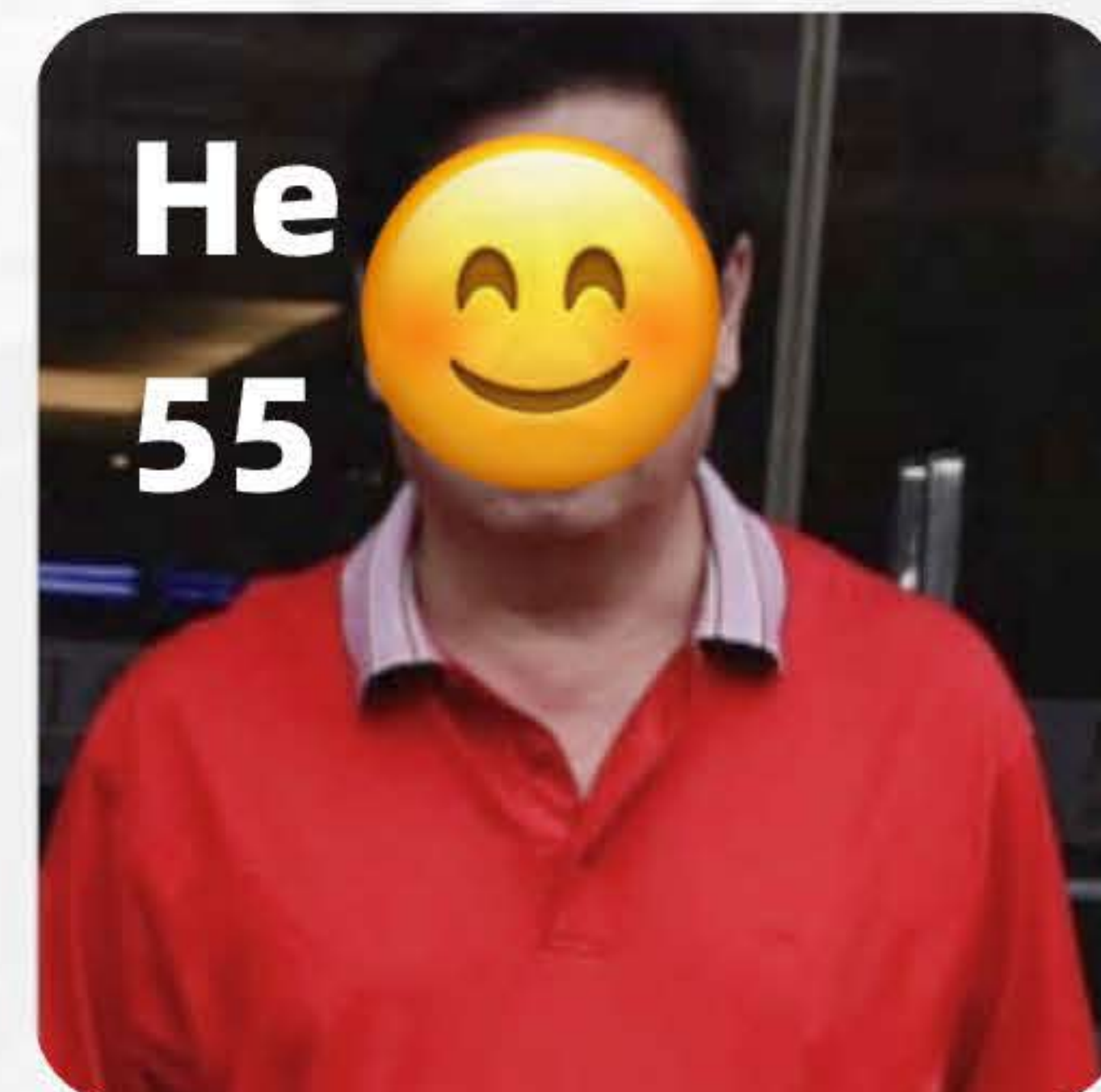
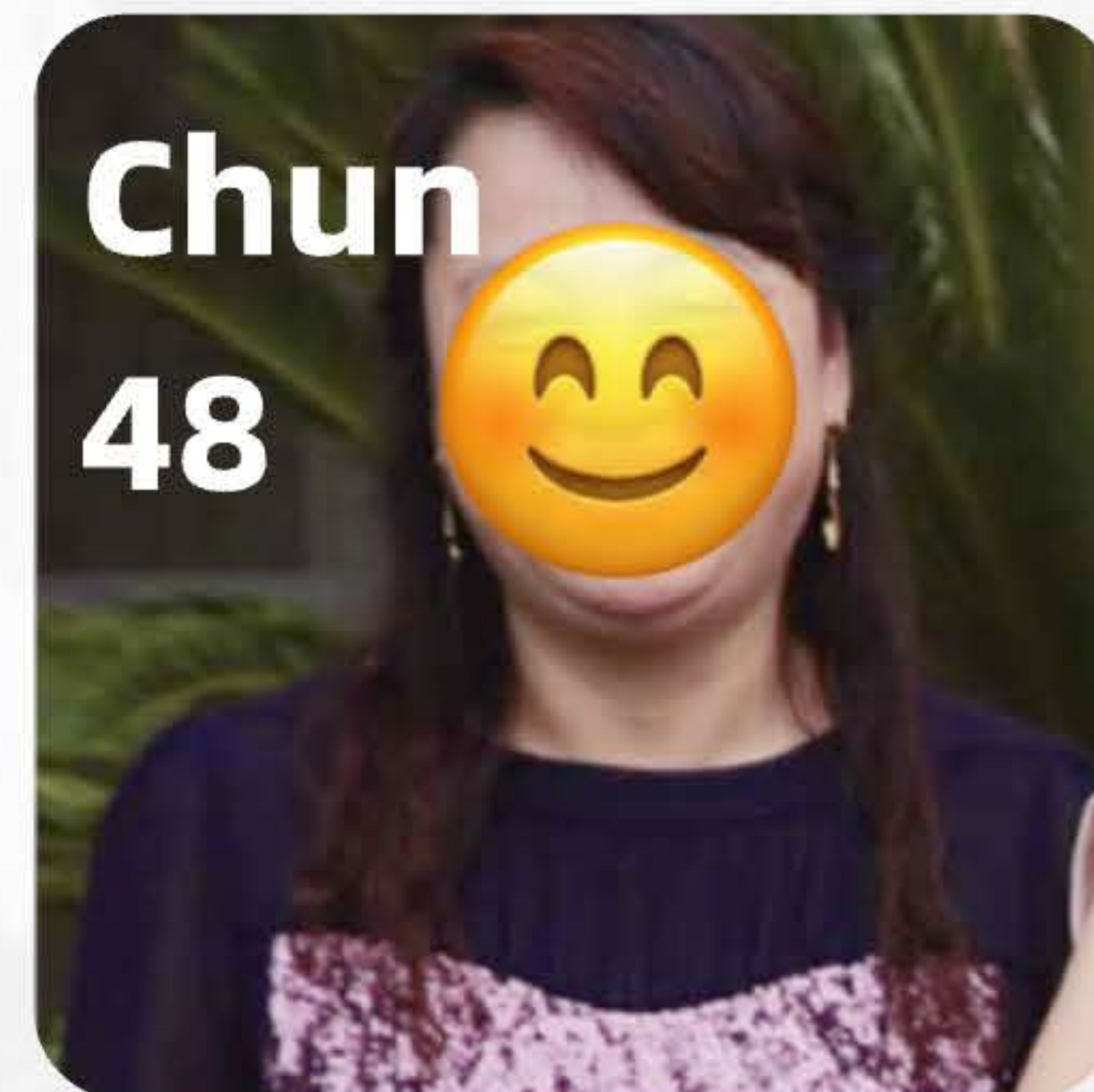
Why A.O smiths' worker complain?

2. Focus Group & Co-design Workshop

5 Participants

90 Minutes

3 Q & A s



Q: We didn't provide the landscape interface for BL3.0, Why did you put it on the PADs?

A: Larger font size and bigger buttons for easier viewing and clicking;

Q: Why do you find the operation cumbersome

A: Recording material input and work reporting data requires clicking at least 10 times or more. The number of pages to navigate through is also excessive.

Q: Draw an ideal operational interface

A:

Users' design language



When the production scenario is **fixed workstation**, mobility is not an advantage. Rather, **readability** is a stronger requirement.

Painpoint & Inspiration

Cyber: BLIM

Clicks

It takes **14 clicks** for a worker to complete one work recording

4

Production Order Selecting

Production Process Selecting



8

Task Selecting

Worker Selecting



2

Recording

Submit



Physical in A.O

Single-person | real-time work reporting

Fixed Workstations

Key insight:

Change the BL3.0 path logic based on workers' real behaviors

Click in



what if:

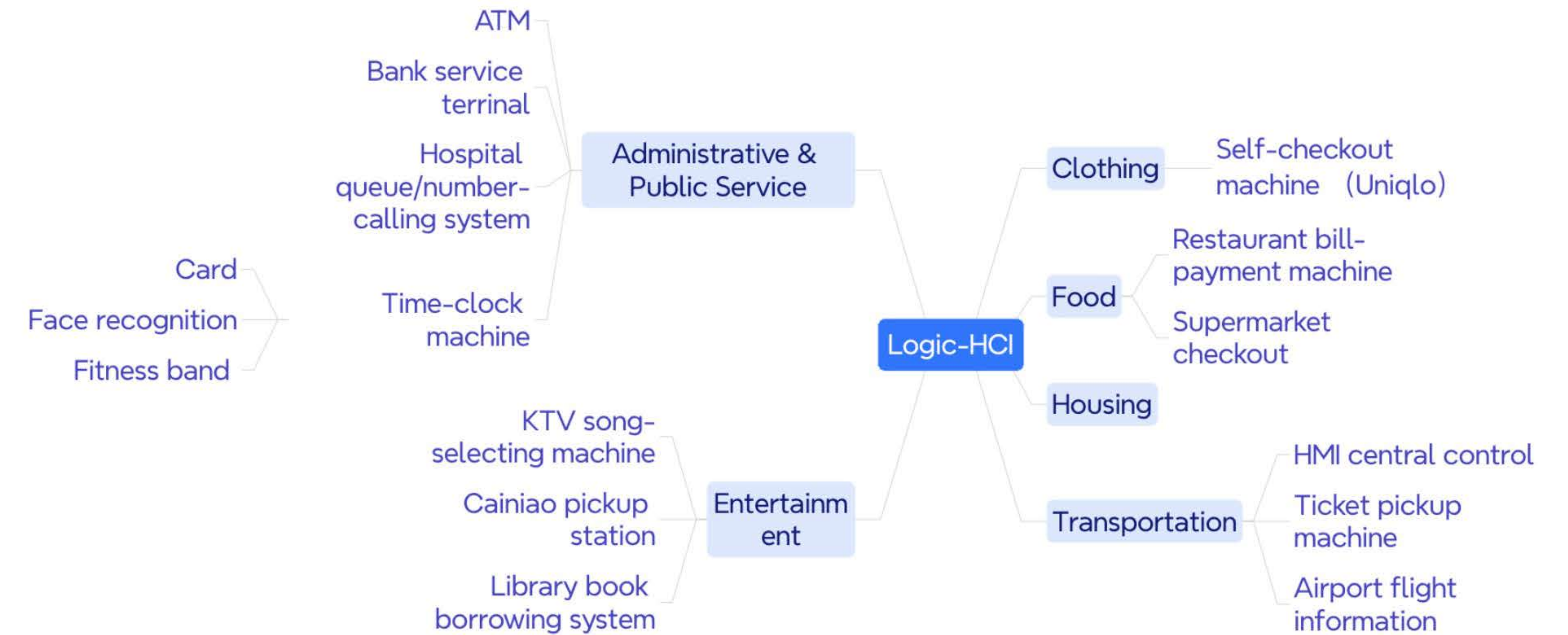


Step1: ~~Select Production Order~~

➡ Scan work ID card

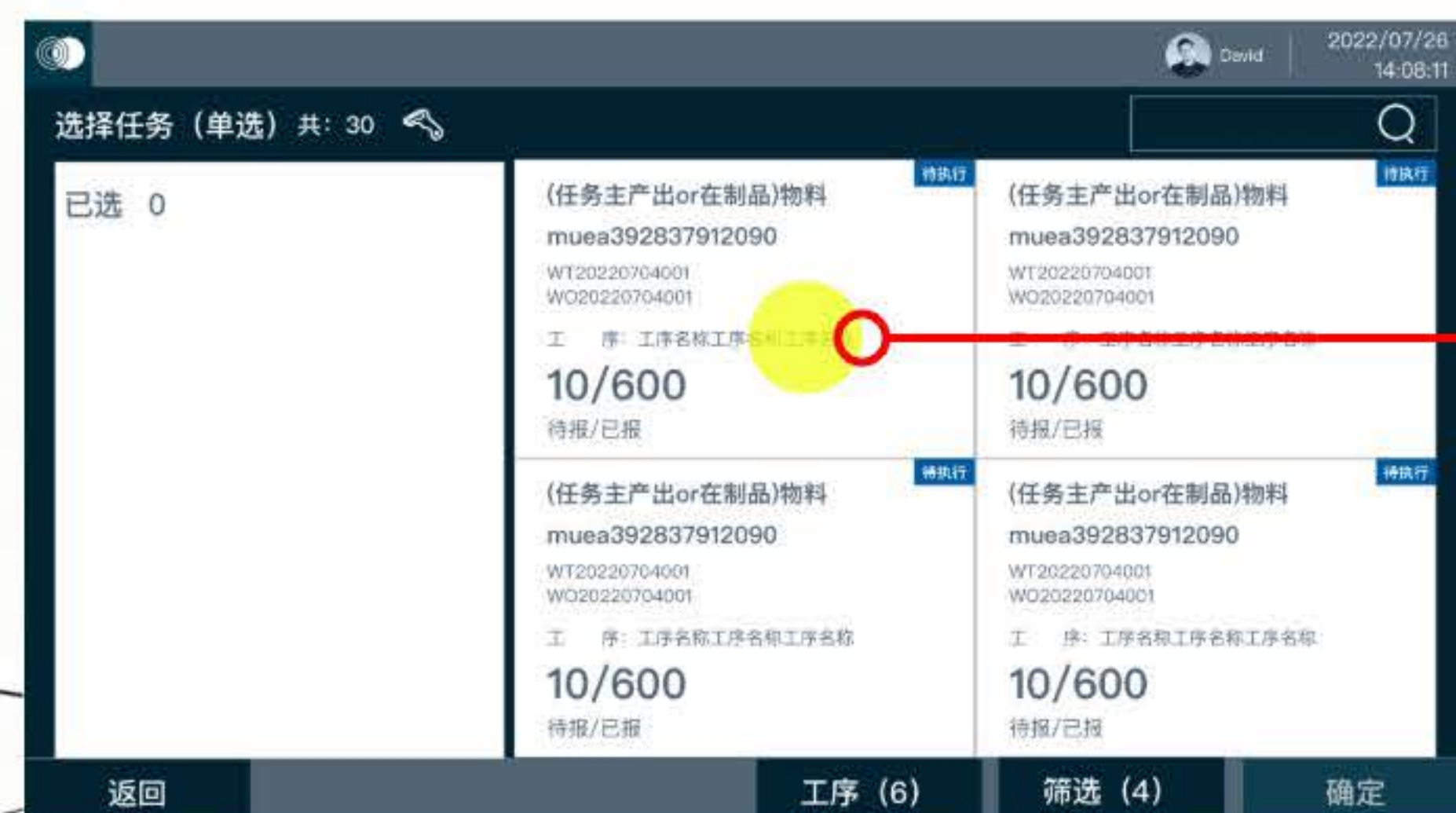
Inspiration & Design

Even in the manufacturing, the operating system lacks competitive products, However, keywords such as landscape mode and simple operation are common.

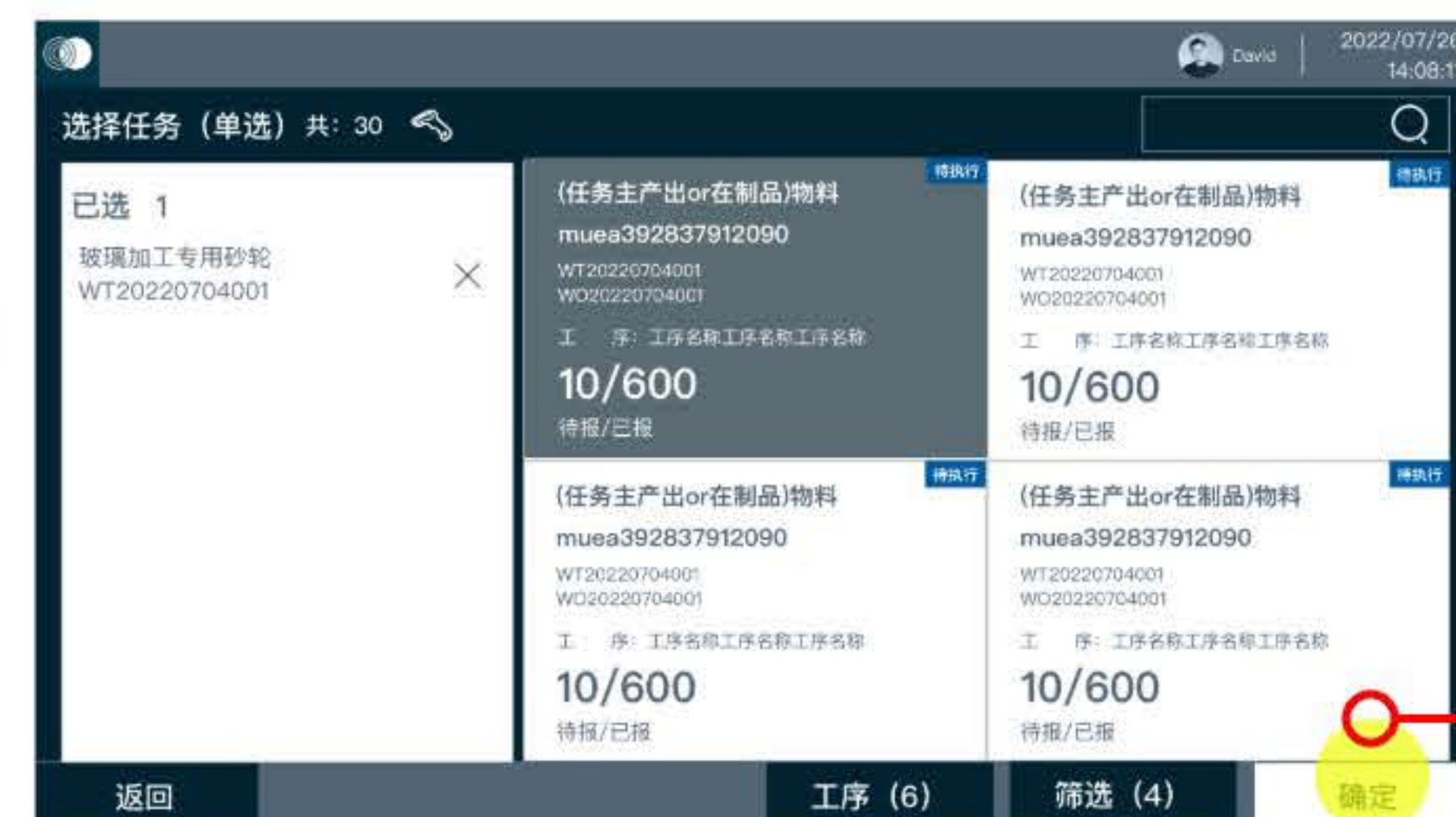


Production scenario eg:

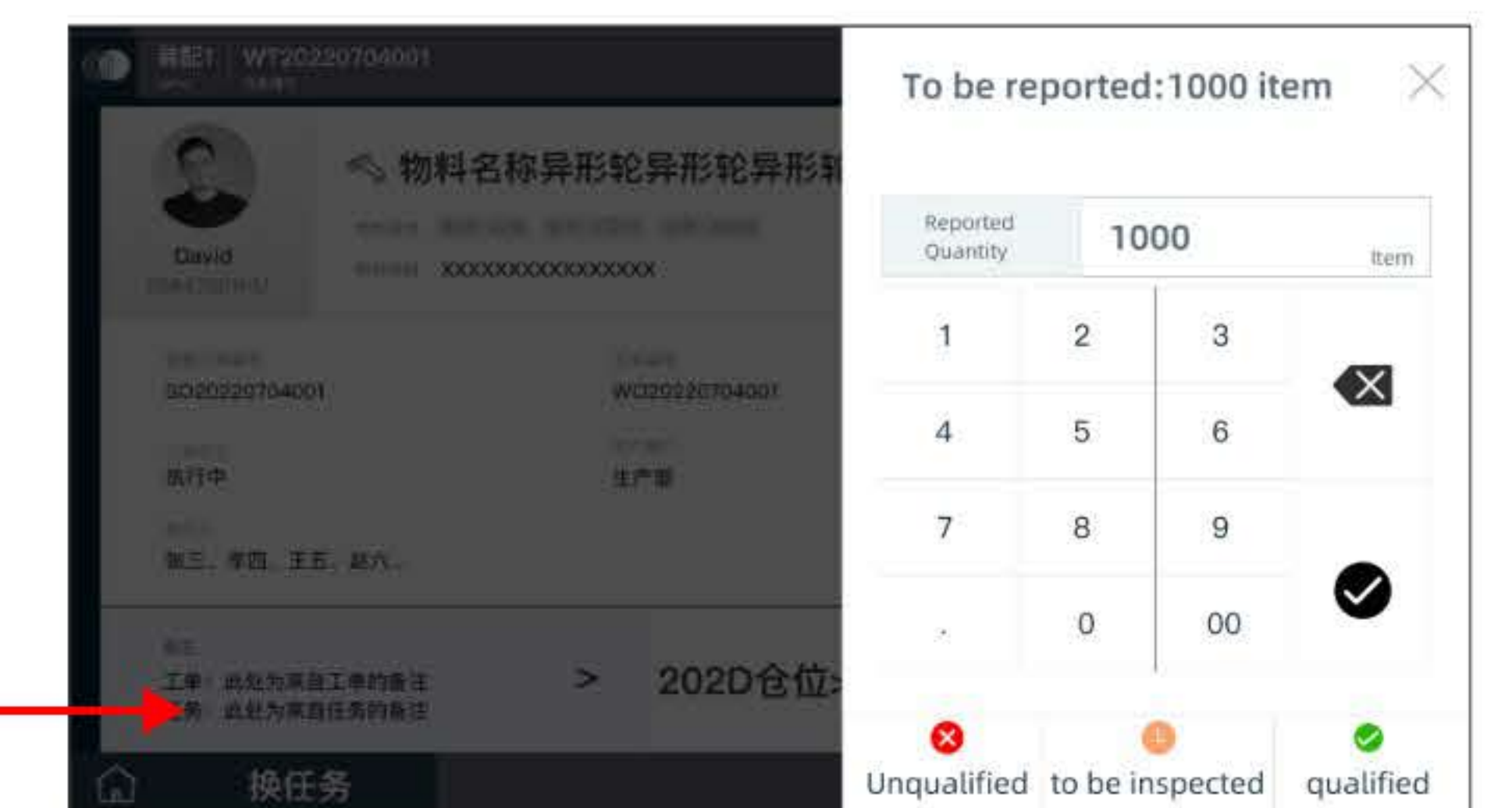
Starts the shift



Selects Task

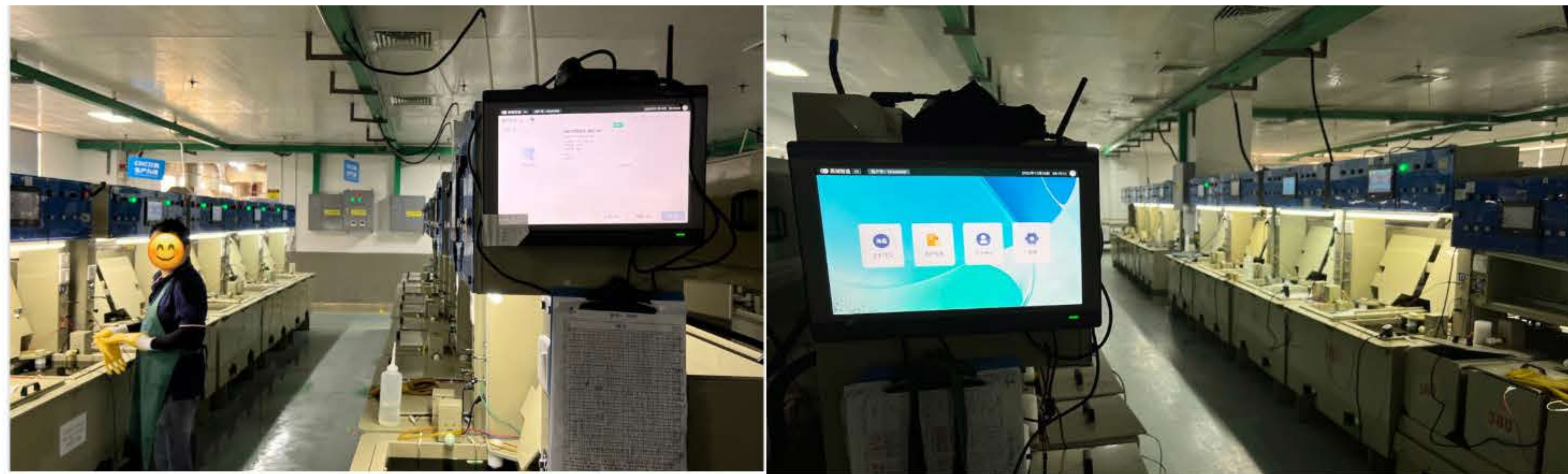


Records Work



And identifies the task that worker A needs to perform

Release & Feedback



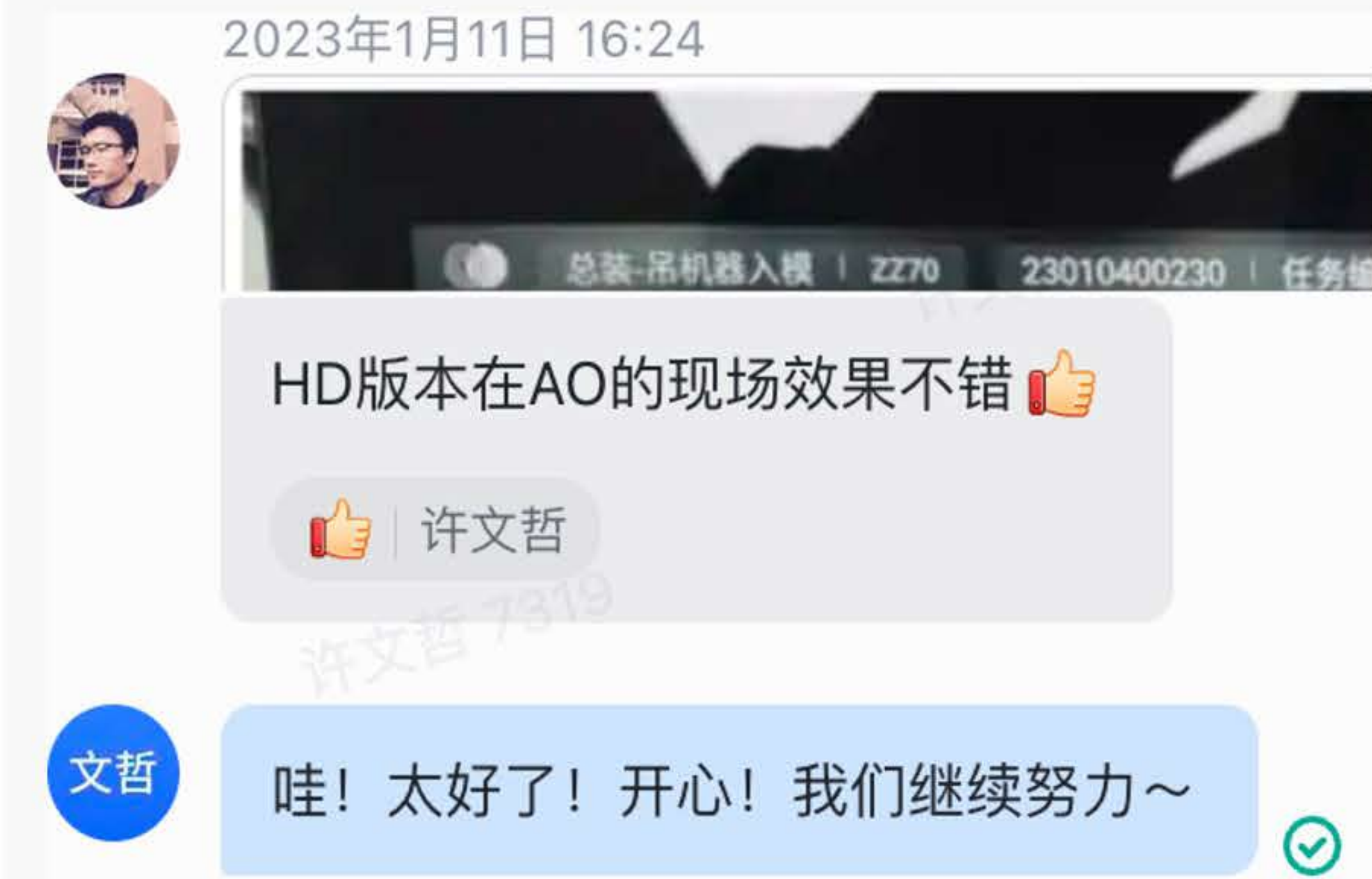
Blacklake-HD (BL-HD) (worker side)

based on BL3.0
Scenario: **Fixed workstation.**
Hardware terminal: **Industrial Pad.**

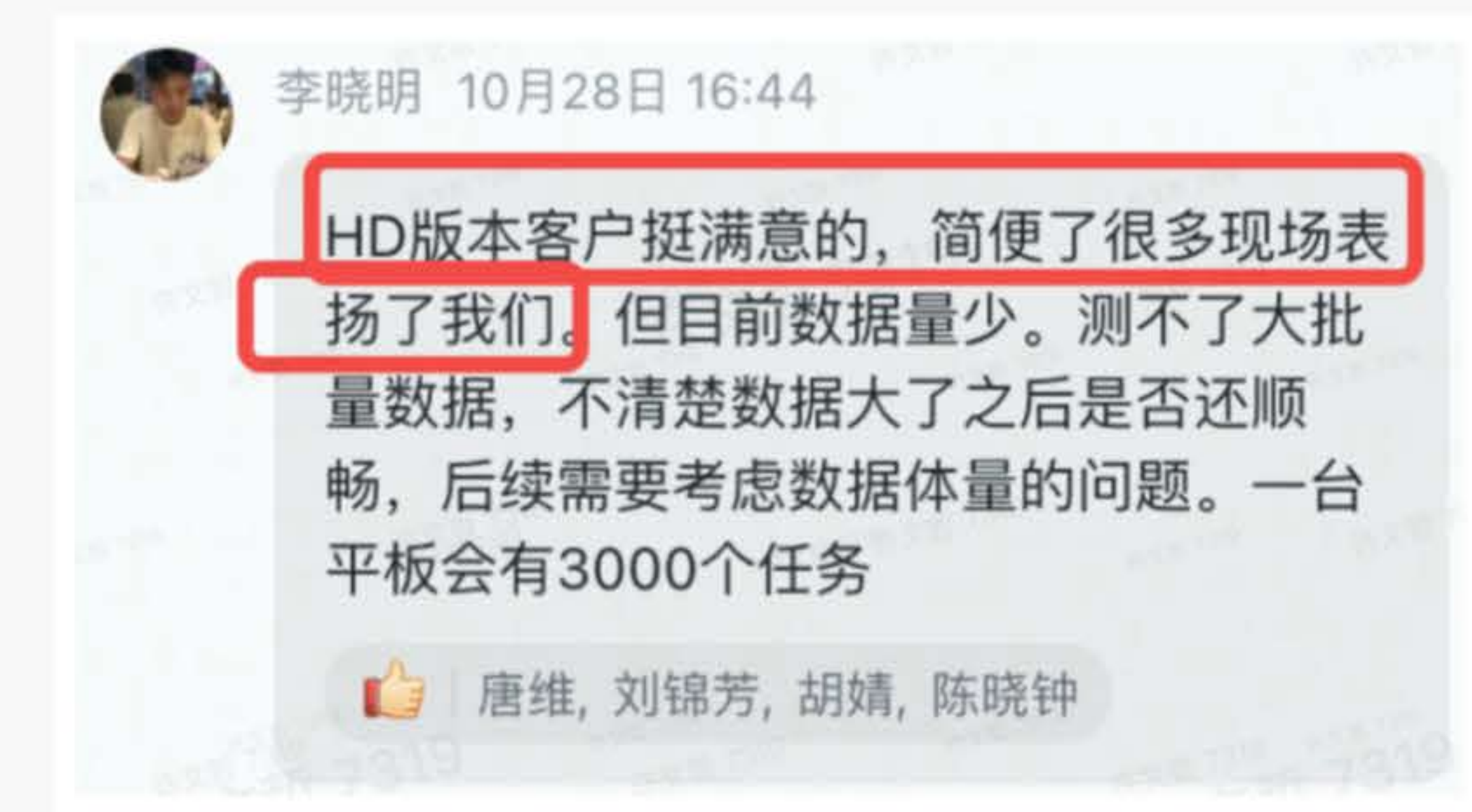
🙏 Thankful for this opportunity. ➡

😊 I made it.

Positive Review from GM



Positive Review from customer



Positive Review from the Sales Director

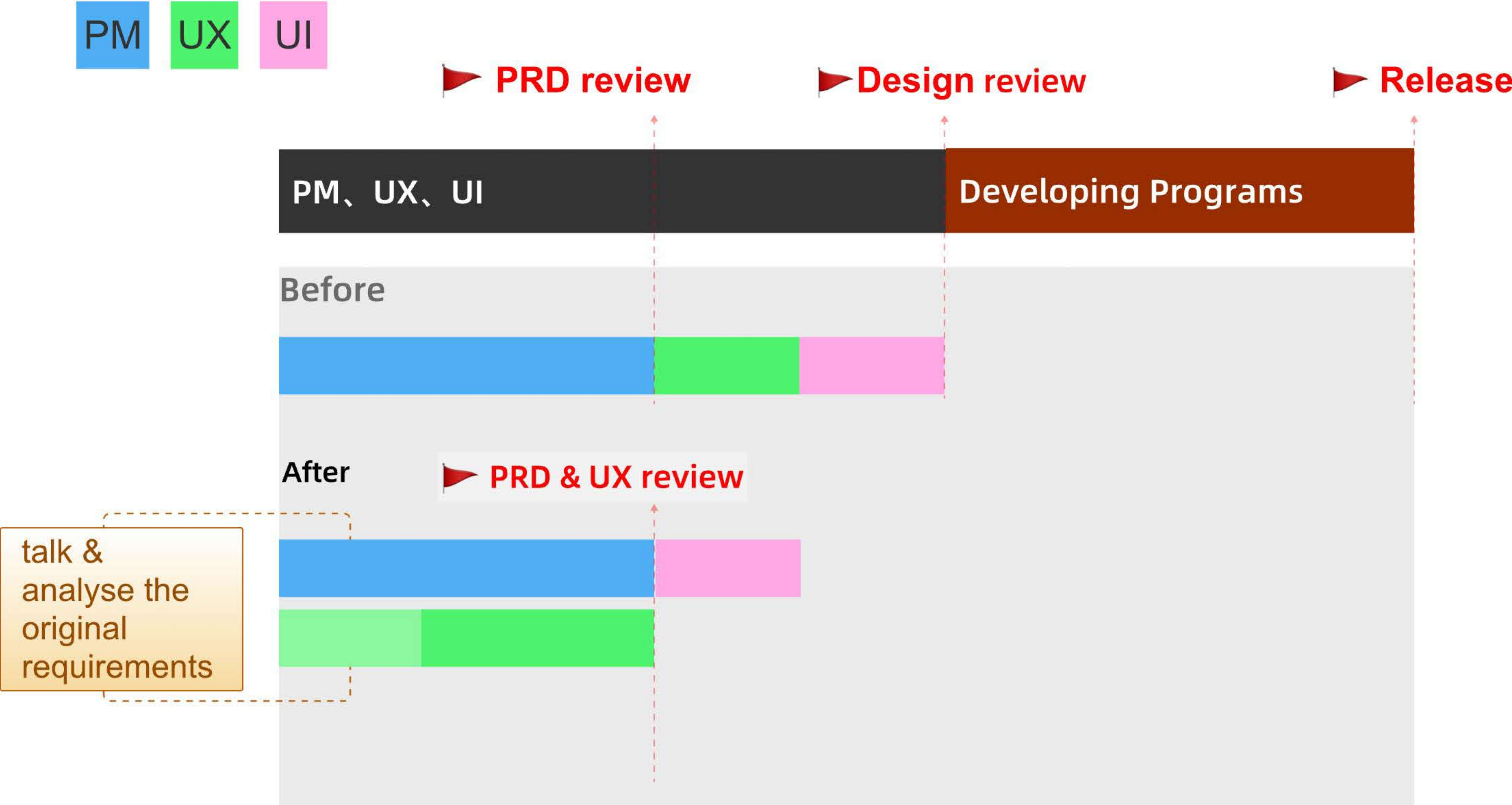


Min
The HD version has a good on-site effect in A.O.

文哲 Me
Wow! Great! happy! That makes me more energy and passion to continue this project~

A new collaboration framework for PM , Designer & Dev.

Clarify and emphasize the value of designers in product functionality implementation and team collaboration



智造产品研发协作流程&各角色职责

说明及宣发

2023年11月28日 (周二)

进入群聊

发起视频会议

VC: 8487870111750

F2-注意

品: 17位参与者

创建会议纪要 或关联已有文档

一、会议背景:
更新智造产品研发流程, 明确各角色职责, 促进产研高效协作

二、会议相关文档: ① 智造产品研发流程-V2.0 (请提前阅读)

三、会议要求: 参会人员为必须参加, 且需要到2023年12月10日

智造产品研发协作流程&各角色职责说明及宣发

文档管理

智造产品研发流程-V2.0

Promote to each team's TLs, PMO created all staff promotion document (only partial excerpt)

节点说明

节点	核心产出	负责人	节点重点事项	参照规范文档
产品方案设计	产品PRD (未定稿)	PM&UX	- PM - 基于产品PRD规范, 完成产品PRD的编写, 提供可供评审的PRD; - UX - 前置参与完整的产品解决方案设计过程, 提供有建设性的交互设计建议;	① 黑湖智造3.0产品需求文档 (PRD) 模板 ② 黑湖智造3.0 产品需求提交规范 ③ 黑湖智造3.0 接口需求提交规范
需求方案评审	产品PRD & UX稿	PM&UX	- PM - 提供可供评审的产品PRD, 截止评审前一个工作日: - 组织研发测试等相关人员进行方案评审, 并进行方案阐述和答疑; - 确认是否需要演示; - 更新需求JIRA状态; - UX - 提供可供评审的UX交互稿, 组织UX评审; - 需求方案评审时, 核心交互设计稿, 需要进行方案阐述和答疑; - 研发 - 参与评审, 评审通过后, 确认技术方案设计&技术方案评审时间, 若有数据清洗, 需提供数据清洗的技术方案; 技术方案评审时间给一个初步时间; - 测试 - 参与方案评审, 并评估是否需要用用例评审; - 在需求方案评审通过后, 根据PRD的交互稿提供对应的UI设计	④ 需求评审流程规范

For UX-focused requirements, PMs should proactively communicate with designers in advance.